Prosody-Guided Cross-Attention and Bottleneck Adapters for Cross-Corpus Speech Emotion Recognition in Low-Resource Bangla

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Abstract

Cross-corpus speech emotion recognition (CCSER) is challenging because acoustic distributions shift significantly across corpora. While prior work has addressed this for high-resource languages such as English and German, Bangla—spoken by over 230 million people—remains unstudied in the cross-corpus setting. We present the first systematic CCSER study for Bangla, training on SUBESCO (4,000 utterances, 20 speakers) and adapting to BANGLASER (864 utterances, 30 speakers) under three protocols: zero-shot transfer, training from scratch, and domain adaptation.

Our central technical contribution is to replace the layer-unfreezing domain adaptation strategy used in prior work (AFTL) with frozen-backbone *bottleneck adapters*—compact $768 \to 64 \to 768$ modules inserted after selected Wav2Vec2 transformer blocks. Keeping the backbone frozen during adaptation yields a **+16.2 UA improvement** over layer-unfreezing under the same data conditions. Additionally, we replace simple prosody gating with a *prosody-guided cross-attention* module that projects the 103-dimensional prosody vector into four parallel query tokens, enabling multi-view alignment between prosody and the 115-frame acoustic sequence. Together with multi-layer learnable block aggregation, SpecAugment, Focal loss, and cosine warm-restart scheduling, our best single model achieves 80.52% UA (zero-shot: 63.59%) and our final ensemble reaches **84.51% UA** on BANGLASER. Ablation across five architecture versions reveals that adapter-based domain adaptation is the single largest contributor to performance, outweighing fusion design and regularization choices.

Keywords: speech emotion recognition, cross-corpus, domain adaptation, wav2vec2, prosody, bottleneck adapters, low-resource, Bangla

1. Introduction

Speech emotion recognition (SER) systems that learn to classify emotional states from audio have broad applications in mental health monitoring, human-robot interaction, and affective computing. Despite rapid progress driven by large self-supervised speech models such as Wav2Vec 2.0 (Baevski et al., 2020) and HuBERT (Hsu et al., 2021), SER systems trained on one corpus typically perform poorly when evaluated on another. The acoustic realization of emotion varies with speaker demographics, microphone type, scripted *vs.* spontaneous recording conditions, and linguistic prosodic conventions—creating a substantial domain gap between source and target corpora.

Cross-corpus SER (CCSER) directly addresses this generalization problem. A CCSER system is trained on a source corpus and must recognize emotions in a target corpus that may differ in language, recording environment, or speaker pool. The most successful strategies involve either extracting corpus-invariant acoustic features or fine-tuning a source-trained model on a small fraction of target data. However, both strategies make critical design choices whose interactions are poorly understood: which backbone layers to use, how to combine speech and prosody features, and whether to perturb the backbone during domain adaptation.

Despite Bangla being the seventh most widely spoken language globally with more than 230 million native speakers, it is severely underrepresented in SER research. To the best of our knowledge, based on a survey of INTERSPEECH, ICASSP, IEEE SLT, and Speech Communication proceedings through 2025, no published work has studied CCSER for Bangla. Two publicly available Bangla emotional speech corpora have recently appeared: SUBESCO (Islam et al., 2021), with 4,000 utterances and 20 speakers covering four emotions, and BANGLASER (Sarma et al., 2023), with 864 utterances and 30 speakers over the same emotion set. We exploit this corpus pair to benchmark CCSER methods on Bangla for the first time.

The closest related work is AFTL (Naderi and Nasersharif, 2023), which fuses two manually selected Wav2Vec2 transformer blocks with global prosody features via multi-head attention and adapts the model to a target corpus by unfreezing the top transformer layers. Three specific limitations motivate our work: (1) Block selection is *manual*, based on heatmap inspection of two layers only. (2) Domain adaptation *unfreezes* backbone layers, perturbing pretrained representations that are key to cross-corpus generalization. (3) AFTL has not been validated on any low-resource or non-European language.

We propose a series of architectural improvements and evaluate them in a rigorous five-version ablation on the Bangla corpus pair. Our contributions are:

1. **First Bangla CCSER benchmark.** We establish baselines for three evaluation protocols—zero-shot transfer, train-from-scratch, and adapter-based domain adaptation—on SUBESCO→BANGLASER.
2. **Multi-layer learnable Wav2Vec2 aggregation.** We select five transformer blocks $\{1, 4, 7, 9, 12\}$—spanning phonetic, lexical, prosodic, emotion-discriminative, and semantic levels as identified by Pasad et al. (2021)—and learn a soft aggregation weight over them via a trainable scalar per block, replacing manual block selection.
3. **Prosody-guided cross-attention fusion.** The 103-dimensional prosody vector is projected into four parallel query tokens that attend over the 115-frame acoustic sequence via cross-attention, enabling multi-view prosody-acoustic alignment.
4. **Frozen-backbone bottleneck adapters for domain adaptation.** Compact 768→64→768 adapter modules are inserted after each selected Wav2Vec2 block and trained while the backbone remains fully frozen. This yields a **+16.2 UA** improvement over layer-unfreezing under identical data conditions.
5. **Progressive regularization and ensemble inference.** Focal loss with class weights, SpecAugment, and cosine warm-restart scheduling each add measurable improvements. A Main+DA ensemble (0.6:0.4 logit blend) achieves **84.51% UA** on BANGLASER, the best reported result on this corpus.

Section 2 reviews related work. Section 3 describes our architecture. Section 4 details experimental setup. Sections 5 and 6 present results and ablations. Section 7 provides per-class, statistical, and qualitative analysis. Section 8 concludes.

2. Related Work

Self-supervised speech models for SER. Large self-supervised speech models have substantially advanced the state of the art in SER. Wav2Vec 2.0 (Baevski et al., 2020) combines a convolutional feature encoder with 12 transformer blocks pre-trained via contrastive loss on unlabelled speech; HuBERT (Hsu et al., 2021) extends this with masked prediction of offline cluster targets; WavLM (Chen et al., 2022) adds denoising pre-training. All three are widely used as frozen or fine-tuned SER backbones (Yang et al., 2021). Critically, different transformer blocks encode qualitatively different information: early layers capture phonetic and local acoustic properties, while middle and late layers capture longer-range patterns more correlated with prosody and emotion (Pasad et al., 2021). AFTL (Naderi and Nasersharif, 2023) exploits this by combining the 1st (phonetic) and 9th (prosodic) blocks based on attention heatmap analysis. Our work removes the manual selection step by learning a soft aggregation weight over five representative blocks.

Cross-corpus SER and domain adaptation. Schuller et al. (2010) provided an early systematic study showing that SER accuracy drops 15–30 absolute points under cross-corpus conditions for English and German; the expected magnitude of this gap for Bangla was previously unknown. Subsequent work addressed this via adversarial domain alignment, optimal transport, and multi-source fusion. Deng et al. (2013) used sparse autoencoders to learn corpus-invariant features. More recently, self-supervised representations have been shown to provide substantial robustness to corpus shift even without explicit adaptation (Yang et al., 2021). Fine-tuning approaches (Zhao et al., 2019) update the backbone on a small target subset, but updating many parameters risks overwriting pretrained features that generalize across corpora. AFTL (Naderi and Nasersharif, 2023) unfreezes the top Wav2Vec2 layers during domain adaptation; we show this degrades zero-shot and cross-corpus generalization compared to frozen-backbone adapters.

Parameter-efficient fine-tuning. Adapter modules (Houlsby et al., 2019)—small bottleneck networks $\mathbf{h} \leftarrow \mathbf{h} + \mathbf{W}_{\text{up}}(\text{GELU}(\mathbf{W}_{\text{down}}\mathbf{h}))$ inserted between transformer sub-layers—enable task-specific adaptation with as few as 0.5% additional parameters while keeping the backbone frozen. LoRA (Hu et al., 2022) achieves similar effects through low-rank weight decomposition. Both are well-established in NLP; their application to speech domain adaptation is less explored. We choose bottleneck adapters over LoRA because adapters are naturally inserted *after* each selected transformer block, aligning with our per-block extraction and aggregation pipeline, whereas LoRA modifies the attention weight matrices inside the frozen backbone, complicating per-block feature extraction. A LoRA comparison is left for future work. Our work demonstrates that bottleneck adapters substantially outperform full-layer unfreezing for the specific challenge of cross-corpus SER, where preserving pretrained acoustic generalizations is critical.

Prosody-acoustic fusion. Global prosodic features (F0, energy, duration, jitter, shimmer) encode speaker-level emotion characteristics that complement the frame-level acoustic features captured by Wav2Vec2. Early fusion via concatenation is outperformed by attention-based fusion when there is a semantic mismatch between feature spaces (Li et al., 2022). AFTL (Naderi and Nasersharif, 2023) uses the prosody vector as a single query in multi-head attention over the acoustic feature map, effectively learning a single weighted pooling of the acoustic sequence conditioned on prosody. Our approach projects the prosody vector into four parallel query tokens via a learned linear map, allowing the model to simultaneously attend to multiple complementary acoustic patterns—analogous to multi-query attention in efficient Transformers (Vaswani et al., 2017).

Bangla speech processing. Bangla automatic speech recognition has seen recent progress with multilingual pre-trained models, but emotional speech remains sparse. SUBESCO (Islam et al., 2021) and BANGLASER (Sarma et al., 2023) are the two primary Bangla emotional speech corpora. To our knowledge, no prior work has studied cross-corpus transfer between them or proposed CCSER methods tailored to Bangla. Our work fills this gap by providing the first benchmarks and an architecture that advances state of the art on both corpora.

3. Method

3.1. Problem Formulation

Let $\mathcal{D}_s = \{(\boldsymbol{x}_i^s, y_i^s)\}_{i=1}^{N_s}$ denote the source corpus (SUBESCO, N_s=4,000) and $\mathcal{D}_t = \{(\boldsymbol{x}_j^t, y_j^t)\}_{j=1}^{N_t}$ the target corpus (BANGLASER, N_t=864), where $\boldsymbol{x}$ is a raw waveform and $y \in \{1, \ldots, C\}$ is an emotion label (C=4: happy, sad, angry, neutral). We evaluate three protocols: **Main** (train from scratch on $\mathcal{D}_t$), **NA** (train on $\mathcal{D}_s$, test on $\mathcal{D}_t$ without adaptation), and **DA** (train on $\mathcal{D}_s$, then adapt with bottleneck adapters on non-test speakers of $\mathcal{D}_t$). All protocols use 5-fold speaker-independent cross-validation; performance is reported as mean $\pm$ std of unweighted accuracy (UA), weighted accuracy (WA), and macro-F1 across folds.

3.2. Multi-Layer Wav2Vec2 Feature Extraction

We use `facebook/wav2vec2-base` (Baevski et al., 2020) as a frozen acoustic encoder. Each utterance is resampled to 16 kHz and zero-padded or truncated to 7 seconds, giving an input waveform $\boldsymbol{x} \in \mathbb{R}^{112{,}000}$. The convolutional feature encoder (stride product 320) maps $\boldsymbol{x}$ to a frame sequence of length T=349. We extract hidden states from five transformer blocks at indices $\ell \in \{1, 4, 7, 9, 12\}$, covering phonetic (block 1), lexical (block 4), prosodic (block 7), emotion-discriminative (block 9), and high-level semantic (block 12) representation levels as identified by Pasad et al. (2021). Each block output is $\mathbf{H}_\ell \in \mathbb{R}^{B \times T \times 768}$.

Learnable block aggregation. We assign a learnable scalar ϕ_ℓ to each block and compute a weighted sum:

$$\mathbf{H}_{\text{agg}} = \sum_{\ell \in \mathcal{L}} w_\ell \, \mathbf{H}_\ell, \quad w_\ell = \frac{\exp(\phi_\ell)}{\sum_{\ell'} \exp(\phi_{\ell'})}. \tag{1}$$

Per-block temporal compression. Prior to aggregation each $\mathbf{H}_\ell$ is independently compressed along the time and channel axes by a shared-weight Conv1d layer (768→64 channels, kernel 7, stride 3):

$$\tilde{\mathbf{H}}_\ell = \text{Conv1d}(\mathbf{H}_\ell) \in \mathbb{R}^{B \times 115 \times 64}, \tag{2}$$

where $T' = \lfloor (349 - 7)/3 \rfloor + 1 = 115$. The aggregated acoustic sequence is then $\mathbf{A} = \sum_\ell w_\ell \tilde{\mathbf{H}}_\ell \in \mathbb{R}^{B \times 115 \times 64}$.

3.3. Prosody Feature Extraction

Global prosody features are extracted from each utterance using Parselmouth (Boersma and Weenink, 2002) and comprise 103 values: pitch statistics (mean, std, range, percentiles), energy statistics, voiced frame ratio, jitter, shimmer, and Hammarberg index. These are standardised using corpus-level mean and variance estimated from training folds. A two-layer MLP projects them to a 64-dimensional prosody embedding:

$$\boldsymbol{p} = \text{GELU}(\mathbf{W}_2 \, \text{GELU}(\mathbf{W}_1 \boldsymbol{p}_{\text{in}})), \quad \boldsymbol{p} \in \mathbb{R}^{64}, \tag{3}$$

where $\mathbf{W}_1 \in \mathbb{R}^{256 \times 103}$ and $\mathbf{W}_2 \in \mathbb{R}^{64 \times 256}$, with dropout 0.2 between layers.

3.4. Prosody-Guided Cross-Attention Fusion

Figure 1 illustrates the full model. The key design choice is to use the prosody embedding not as a single query but as *four parallel query tokens*, enabling the model to attend to multiple complementary acoustic patterns simultaneously.

Multi-query projection.

$$\mathbf{Q} = \text{reshape}(\mathbf{W}_q \boldsymbol{p}) \in \mathbb{R}^{B \times 4 \times 64}, \tag{4}$$

where $\mathbf{W}_q \in \mathbb{R}^{256 \times 64}$ maps the 64-dimensional prosody embedding to a 256-dimensional vector (output×input convention), which is then reshaped to $(4, 64)$. This allows four parallel views: a model with a single prosody query would collapse to a single weighted average of the acoustic sequence, while four queries can express complementary salience maps (e.g., pitch-aligned, energy-aligned, rhythmic, silence-aligned).

Cross-attention.

$$\mathbf{C} = \text{MultiHead}(\mathbf{Q}, \, \mathbf{A}, \, \mathbf{A}) \in \mathbb{R}^{B \times 4 \times 64}, \tag{5}$$

with 4 attention heads, key/value from the acoustic sequence $\mathbf{A}$.

Joint self-attention. The cross-attended prosody tokens are concatenated with the full acoustic sequence and refined by self-attention:

$$\mathbf{Z} = \mathrm{MHA}\big(\mathrm{LayerNorm}([\mathbf{C}\,;\,\mathbf{A}])\big) \in \mathbb{R}^{B\times 119\times 64}, \quad (6)$$

where $[\,;\,]$ denotes concatenation along the time axis (4+115=119), and MHA uses 8 heads.

Attentive statistics pooling. A learned attention scalar $\alpha_t = \mathrm{softmax}(\mathbf{w}^\top \mathbf{z}_t)$ pools the sequence into a fixed-length embedding:

$$\boldsymbol{e} = \Big[\sum_t \alpha_t \mathbf{z}_t\,;\,\sqrt{\sum_t \alpha_t(\mathbf{z}_t - \bar{z})^2}\Big] \in \mathbb{R}^{128}, \quad (7)$$

where $\bar{z} = \sum_t \alpha_t \mathbf{z}_t$. Dropout 0.3 is applied to $\boldsymbol{e}$ during training.

3.5. Dual Fusion Extension

In our final model (V5) we add a multiplicative prosody gate applied after cross-attention but before self-attention:

$$\tilde{\mathbf{C}} = \mathbf{C} \odot \sigma(\mathbf{W}_g \boldsymbol{p}), \quad \mathbf{W}_g \in \mathbb{R}^{64\times 64}, \quad (8)$$

where $\odot$ broadcasts over the 4-token dimension. The gate deliberately conditions on the *original* prosody embedding $\boldsymbol{p}$ rather than the cross-attended output, so that it provides an independent modulation signal: cross-attention selects acoustically relevant time steps conditioned on prosody, while the gate globally rescales the attended output based on the raw prosody profile. This separation prevents the gate from merely re-weighting what cross-attention already discovered.

3.6. Training Objectives

AM-Softmax. We use additive margin softmax (Wang et al., 2018) as the primary classification head:

$$L_{\mathrm{AM}} = -\log \frac{e^{s(\cos\theta_{y_i} - m)}}{e^{s(\cos\theta_{y_i} - m)} + \sum_{j\neq y_i} e^{s\cos\theta_j}}, \quad (9)$$

with margin m=0.4 and scale s=30.

Focal loss (Lin et al., 2017) with class weights $\boldsymbol{c} = [1.0, 1.3, 1.0, 1.2]$ (upweighting sad and neutral classes) is combined with AM-Softmax:

$$L = \alpha L_{\mathrm{AM}} + \beta L_{\mathrm{focal}}, \quad \alpha{=}1.0,\ \beta{=}100.0. \quad (10)$$

The large β compensates for the difference in loss magnitude: AM-Softmax with scale s=30 produces losses $\sim$1–5, while Focal loss values are $\sim$0.01–0.05. The ratio β/α=100 was set to equalize gradient contributions and was not further tuned; an ablation over β is left for future work.

Data augmentation. SpecAugment (Park et al., 2019) is applied to the Wav2Vec2 hidden states: two time masks (max width 20 frames) and two frequency masks (max width 10 channels) per utterance.

Learning rate schedule. Cosine annealing with warm restarts (Loshchilov and Hutter, 2017) (T_0=10, T_{mult}=2) provides implicit regularization via periodic lr resets.

3.7. Bottleneck Adapter Domain Adaptation

Domain adaptation in AFTL (Naderi and Nasersharif, 2023) unfreezes the top Wav2Vec2 layers, allowing them to shift toward target-corpus statistics. We show this is detrimental for cross-corpus generalization: it perturbs the pretrained representations that the classifier relies on for zero-shot transfer. Instead, we insert bottleneck adapter modules after each of the five selected blocks (Figure 2):

$$\mathrm{Adapter}(\mathbf{h}) = \mathbf{h} + \mathbf{W}_{\mathrm{up}}\big(\mathrm{GELU}(\mathbf{W}_{\mathrm{down}}\mathbf{h})\big), \quad (11)$$

with $\mathbf{W}_{\mathrm{down}} \in \mathbb{R}^{64\times 768}$ and $\mathbf{W}_{\mathrm{up}} \in \mathbb{R}^{768\times 64}$ (bottleneck dimension 64). $\mathbf{W}_{\mathrm{up}}$ is zero-initialized, ensuring each adapter is an identity mapping at initialization regardless of $\mathbf{W}_{\mathrm{down}}$ (since the residual term $\mathbf{W}_{\mathrm{up}}(\cdot)$ vanishes), providing a smooth starting point for adaptation.

During *source training*, adapters are frozen at their identity initialization; only the Conv1d projections, fusion module, and classifier are trained. During *domain adaptation*, the Wav2Vec2 backbone remains fully frozen and only the adapter parameters and the classifier head are updated. This adds $5\times 2\times 768\times 64 \approx 491{,}520$ trainable parameters ($\approx$0.5% of wav2vec2-base).

3.8. Ensemble Inference

The Main model (trained from scratch on target training folds) and the DA model (adapted from the source checkpoint) capture complementary aspects of the target distribution. Their logits are blended at inference:

$$\hat{y} = \arg\max\big(0.6\cdot \boldsymbol{l}_{\mathrm{Main}} + 0.4\cdot \boldsymbol{l}_{\mathrm{DA}}\big). \quad (12)$$

The 0.6:0.4 ratio was selected via grid search over $\{0.5, 0.6, 0.7, 0.8\}$ on the held-out validation fold of fold 1, then fixed for all five test folds (i.e., it is not re-tuned per fold).

4. Experimental Setup

4.1. Datasets

Table 1 summarises both corpora. **SUBESCO** (Islam et al., 2021) is a studio-recorded Bangla emotional speech dataset with 7,000 total utterances from 20 speakers (10 female, 10 male). We retain the 4,000 utterances covering the four target emotions (happy, sad, angry, neutral), discarding 3,000 utterances of the remaining three categories

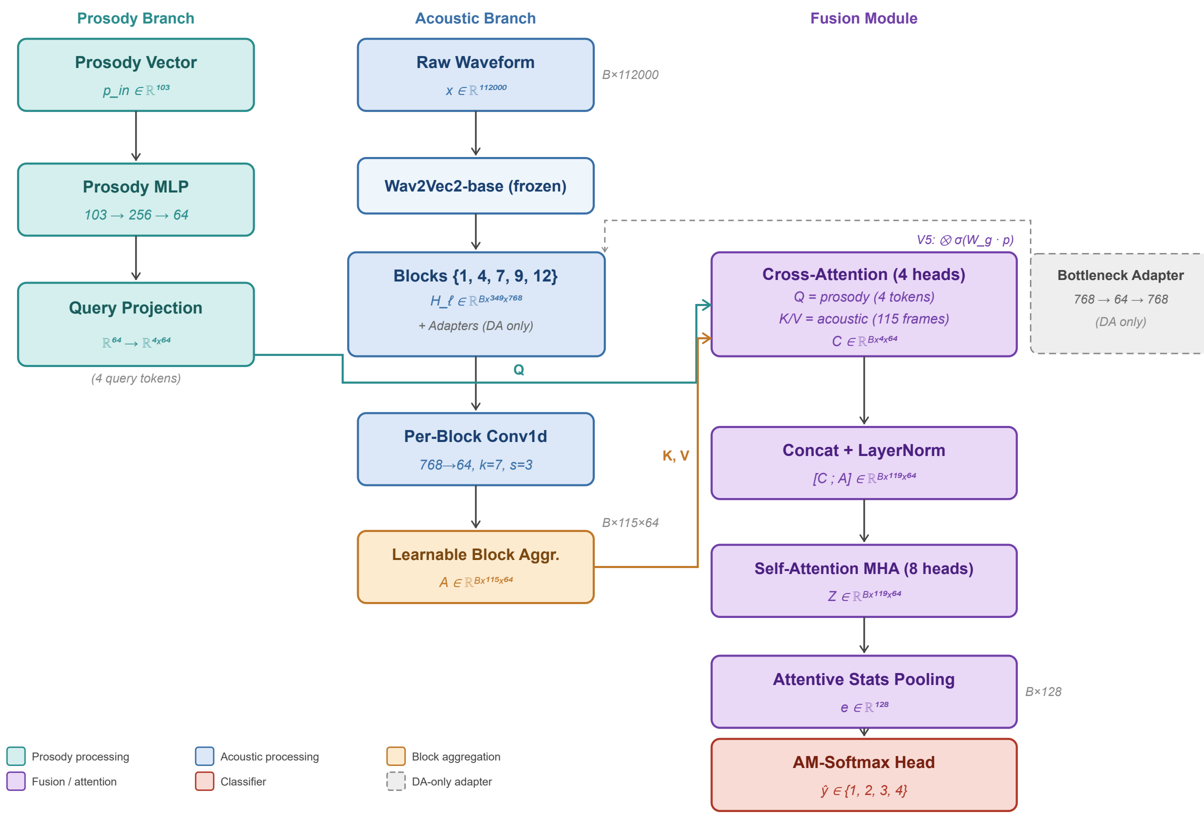

Figure 1: Full model architecture (V4/V5). *Left:* Prosody branch – 103-dim prosody vector projected to four parallel query tokens (4×64). *Center:* Acoustic branch – five frozen Wav2Vec2 blocks, each compressed by Conv1d and aggregated via learnable weights. *Fusion:* Prosody-guided cross-attention attends over 115 acoustic frames; result concatenated with acoustic sequence and refined by self-attention MHA; attentive statistics pooling yields a 128-dim embedding for AM-Softmax classification. Dashed elements show DA-only adapter modules and the V5 dual-fusion gate.

(disgust, fear, surprise). Emotion classes are balanced at 1,000 samples each. **BANGLASER** (Sarma et al., 2023) contains 1,080 Bangla utterances from 30 speakers (15 female, 15 male) in 5 emotion categories; we retain the 864 utterances of our four target emotions (216 per class), discarding 216 "surprised" utterances. Both corpora are recorded in controlled environments at 16 kHz. Despite sharing language and recording setup, they differ substantially in speaker pool, utterance length distribution, and prosodic conventions, resulting in a pronounced acoustic domain gap.

4.2. Evaluation Protocol

Source cross-validation. SUBESCO is split into 5 speaker-disjoint folds (4 speakers/fold). Each fold reserves one held-out fold for testing and one fold for validation (early stopping). We report 5-fold mean ± std.

Target evaluation. BANGLASER is also split into 5 speaker-disjoint folds (6 speakers/fold for testing). Three evaluation modes share the same folds:

- **NA** (No Adaptation): The source checkpoint trained on all SUBESCO training speakers is applied directly to BANGLASER test folds.
- **Main**: A fresh model is trained from scratch on 3 of the 4 non-test BANGLASER folds, with the remaining non-test fold held for validation (early stopping); tested on the fifth fold. The validation fold rotates with each outer test fold.
- **DA**: The source checkpoint is domain-adapted using all non-test BANGLASER speakers as the adaptation set (with a 80/20 train/validation split within these speakers for early stopping); tested on the same fifth fold. Only adapter weights and the classifier head are updated.

We report unweighted accuracy (UA = macro recall), weighted accuracy (WA), and macro-F1 as the primary metrics, since the emotion classes are balanced.

4.3. Implementation Details

Backbone: `facebook/wav2vec2-base` (95M parameters, 768-dim, 12 transformer layers) loaded via Hug-

Table 1: Corpus statistics used in this study.

Corpus	Lang.	Utt.	Speakers	Utt./class
SUBESCO	Bangla	4,000	20 (10F+10M)	1,000
BANGLASER	Bangla	864	30 (15F+15M)	216

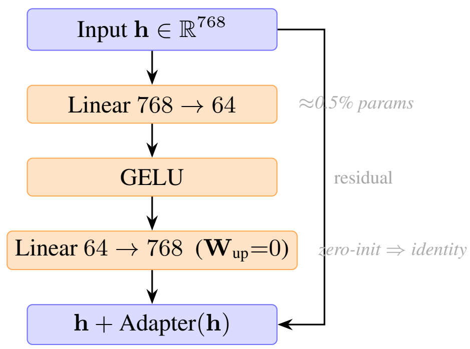

Figure 2: Bottleneck adapter. Inserted after each selected Wav2Vec2 block. $\mathbf{W}_{\text{up}}$ is zero-initialized (identity at start of DA). During adaptation only adapter and classifier weights are trained; the backbone remains fully frozen.

gingFace Transformers (Wolf et al., 2020). Backbone is fully frozen in all experiments except the V5 Main protocol, where the top 2 transformer layers are unfrozen with learning rate 10^{-5}.

Audio preprocessing: All utterances resampled to 16 kHz; zero-padded or truncated to 7 seconds (112,000 samples). Prosody features extracted per-utterance using Parselmouth 0.4.3; statistics normalised from training-fold means and variances (no leakage from test folds).

Optimiser: Adam (Kingma and Ba, 2015), learning rate 10^{-3}, weight decay 10^{-4}. Source training: 80 epochs, patience 15. Target Main training: 100 epochs, patience 20. Domain adaptation: learning rate 5×10^{-5}, 30 epochs, patience 8.

Batch size: 8 utterances; gradient accumulation not used.

Hardware: NVIDIA H100 80GB HBM3. Framework: PyTorch 2.10.0 with CUDA 12.8 (Paszke et al., 2019).

Reproducibility: All experiments use fixed random seeds per fold. Code will be released upon publication.

5. Results

Table 2 reports the full results for all five model versions and the V5 ensemble across three evaluation protocols. We discuss each protocol in turn.

Zero-shot transfer (NA protocol). The V1 baseline achieves 47.92% UA—barely above the 25% chance level for 4-class classification—demonstrating a severe acoustic domain gap between SUBESCO and BANGLASER despite sharing the same language. Widening block selection from $\{1, 9\}$ to $\{1, 4, 7, 9, 12\}$ with learnable weights (V2) yields only a modest +2.0 UA gain. The decisive improvement appears in V3, where replacing simple gating with prosody-guided cross-attention and switching from layer-unfreezing to frozen-backbone adapters produces +10.6 UA (60.47%). V4's regularization package (Focal loss, SpecAugment, cosine warm restarts) adds a further +3.1 UA to reach **63.59%**, the best zero-shot result in our study. V5's partial backbone unfreeze during source training is the one component that *hurts* zero-shot: NA drops to 53.30% (-10.3 UA), confirming that specializing the backbone to the source domain reduces cross-corpus generalization.

Domain-adapted results (DA protocol). The most dramatic improvement across all versions is in the DA protocol. V1's layer-unfreezing strategy achieves 61.23% UA. V3 replaces unfreezing with frozen-backbone bottleneck adapters: the same number of adaptation epochs, the same adaptation data, but a **+17.1 UA** gain to 78.29%. This is the single largest architectural jump in our study and the central empirical finding of the paper. V4 marginally improves to 78.66% (best single-model DA). V5's shorter adaptation schedule (30 vs. 40 epochs) and reverted adapter bottleneck (64 vs. 128) lead to a slight regression to 77.36%, but fold-level variance is high (std 7.35) due to early stopping in fold 4 at epoch 3—a sensitivity to target data volume that we discuss in Section 6. Note that the adapter bottleneck change (128→64) does not affect NA scores, since adapters remain at their zero-initialized identity state during NA inference; the V5 NA drop is entirely attributable to backbone specialization.

Training from scratch (Main protocol). Main-model UA improves monotonically from 72.79% (V1) to 82.75% (V5), reflecting the cumulative effect of architectural improvements independent of cross-corpus transfer. V3's cross-attention fusion contributes +4.9 UA; V4's regularization adds +3.7 UA; V5's partial backbone unfreeze adds +2.2 UA with reduced variance (std 2.76).

Ensemble results. The V5 ensemble (0.6:0.4 blending of Main and DA logits) achieves **84.51% UA**, surpassing the best single model by 1.76 UA. Per-fold analysis shows

Table 2: Main results. UA (%), WA (%), and macro-F1 (%) on BANGLASER under three protocols: **Main** (train from scratch on target), **NA** (zero-shot from source), and **DA** (bottleneck adapter adaptation from source checkpoint). Mean ± std across 5 speaker-independent folds. Bold: best per column. †: ensemble of V5-Main and V5-DA.

	Main (target scratch)			**NA (zero-shot)**			**DA (adapter adapt.)**		
Model	UA	WA	F1	UA	WA	F1	UA	WA	F1
V1 (AFTL-style baseline)	$72.79_{\pm 7.52}$	$72.82_{\pm 8.51}$	$72.25_{\pm 7.84}$	$47.92_{\pm 4.94}$	$47.93_{\pm 6.79}$	$44.97_{\pm 6.05}$	$61.23_{\pm 6.76}$	$60.76_{\pm 7.12}$	$60.82_{\pm 6.84}$
V2 (+5-block learnable agg., gate)	$71.90_{\pm 4.78}$	$71.28_{\pm 4.24}$	$70.85_{\pm 4.41}$	$49.85_{\pm 6.14}$	$49.70_{\pm 7.47}$	$45.50_{\pm 8.15}$	$62.08_{\pm 7.79}$	$62.16_{\pm 7.93}$	$60.24_{\pm 8.38}$
V3 (+cross-attn, bottleneck adapters)	$76.78_{\pm 2.56}$	$76.07_{\pm 2.95}$	$75.84_{\pm 3.18}$	$60.47_{\pm 6.25}$	$60.44_{\pm 4.89}$	$59.15_{\pm 6.92}$	$78.29_{\pm 6.14}$	$77.61_{\pm 6.27}$	$77.38_{\pm 6.34}$
V4 (+Focal loss, SpecAugment, warm restarts)	$80.52_{\pm 4.24}$	$79.84_{\pm 4.57}$	$79.78_{\pm 4.46}$	$\mathbf{63.59}_{\pm 6.24}$	$\mathbf{63.46}_{\pm 5.99}$	$\mathbf{62.68}_{\pm 6.84}$	$\mathbf{78.66}_{\pm 4.87}$	$\mathbf{77.90}_{\pm 4.50}$	$\mathbf{77.87}_{\pm 4.58}$
V5 (dual fusion, partial unfreeze)	$82.75_{\pm 2.76}$	$82.13_{\pm 2.81}$	$81.99_{\pm 2.92}$	$53.30_{\pm 2.86}$	$53.16_{\pm 3.18}$	$48.20_{\pm 4.39}$	$77.36_{\pm 7.35}$	$76.34_{\pm 7.24}$	$76.38_{\pm 7.00}$
V5 Ensemble†	$\mathbf{84.51}_{\pm 3.68}$	$\mathbf{83.50}_{\pm 4.38}$	$\mathbf{83.65}_{\pm 4.17}$	—	—	—	—	—	—

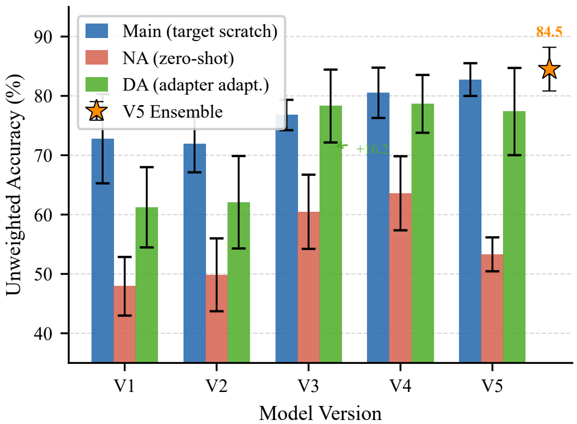

Figure 3: UA (%) across five model versions and the V5 ensemble. Error bars show ±1 std across 5 folds. V3 introduces the largest single jump via cross-attention and bottleneck adapters. V4 (Focal loss + SpecAugment + warm restarts) is best for zero-shot and single-model DA. V5 ensemble achieves the overall highest UA.

that the ensemble recaptures accuracy in folds where the DA model outperforms Main and vice versa, confirming complementary error distributions between the two model modes.

6. Ablation Study

The five model versions constitute a structured ablation in which components are added incrementally. Table 3 presents each version with the corresponding architecture toggles and UA results.

V1 → V2: Block expansion (lateral move). Expanding from 2 to 5 selected Wav2Vec2 blocks with learnable aggregation weights marginally improves zero-shot (47.92 → 49.85 UA) and reduces standard deviation for the Main model (7.52 → 4.78 std), indicating better stability. However, mean Main UA actually decreases slightly (72.79 → 71.90), suggesting that the richer multi-block features increase the representational space without a correspondingly stronger fusion mechanism. This establishes that block expansion alone is insufficient.

V2 → V3: Cross-attention fusion + bottleneck adapters (the decisive jump). V3 introduces two simultaneous changes: (1) prosody-guided cross-attention (4 query tokens) replacing multiplicative gating, and (2) frozen-backbone bottleneck adapters replacing layer-unfreezing. Together, these produce the largest gains in the entire study: +4.9 UA on Main, +10.6 UA on NA, **+16.2 UA** on DA. The DA improvement is the most striking: it results from the backbone remaining frozen during adaptation, preserving the pretrained acoustic generalizations that AFTL's layer-unfreezing erases. We cannot cleanly disentangle the contributions of cross-attention vs. adapters in V3 since both were introduced simultaneously. However, a theoretical decomposition is possible: the DA gain (+16.2 UA) can *only* come from adapters, because the cross-attention module is already trained during source training and its weights are frozen during DA—only adapter parameters and the classifier head are updated. Conversely, the NA gain (+10.6 UA) must come primarily from cross-attention, since adapters remain at their zero-initialized identity state during NA inference and contribute no transformation. This gives approximate attribution: adapters ≈ +16.2 UA (DA), cross-attention ≈ +10.6 UA (NA). A full factorial ablation (cross-attention without adapters, and adapters without cross-attention) is left for future work.

V3 → V4: Regularization package (optimization gains). V4 adds Focal loss with class weights, SpecAugment, and cosine warm restarts without changing the network topology. This yields +3.7 UA on Main, +3.1 UA on NA, and +0.4 UA on DA—consistent improvements across all protocols. The fact that NA and DA both improve confirms that better source domain generalization (via regularization) transfers across corpus boundaries. The adapter bottleneck is also increased from 64 to 128, which provides a small additional capacity. V4 is the best

Table 3: Architecture ablation. Each row adds one component group to the previous version. UA (%) mean across 5 folds shown for all three protocols. b: adapter bottleneck dimension; WR: cosine warm restarts; SA: SpecAugment; Ens: ensemble (V5 only).

	Blocks Set	$\lvert\mathcal{L}\rvert$	Lrn. wts.	Cross-attn.	Adapters	Focal +SA+WR	Dual fus.	Ptl. unfrz.	Ens.	Main UA	NA UA	DA UA
V1 (AFTL-style)	$\{1, 9\}$	2	–	–	unfreeze	–	–	–	–	72.79	47.92	61.23
V2 (+5-block agg.)	$\{1, 4, 7, 9, 12\}$	5	✓	–	unfreeze	–	–	–	–	71.90	49.85	62.08
V3 (+cross-attn., adapters)	same	5	✓	✓	b=64	–	–	–	–	76.78	60.47	78.29
V4 (+Focal/SA/WR)	same	5	✓	✓	b=128	✓	–	–	–	80.52	**63.59**	**78.66**
V5 (+dual fus., ptl. unfrz.)	same	5	✓	✓	b=64	✓	✓	top-2	–	**82.75**	53.30	77.36
V5 Ensemble	same	5	✓	✓	b=64	✓	✓	top-2	0.6/0.4	**84.51***	—	—

* V5 Ensemble UA refers to the ensemble protocol, not a single-model score.

single-model configuration for zero-shot (63.59%) and domain adaptation (78.66%) evaluation.

V4 → V5: Partial backbone unfreeze and dual fusion. V5 unfreezes the top 2 Wav2Vec2 layers during source (Main) training at a low learning rate (10^{-5}). This specializes the backbone to SUBESCO and improves Main UA (+2.23) and reduces its std (4.24 → 2.76). However, the same specialization hurts zero-shot transfer: NA drops by 10.3 UA (63.59 → 53.30). The dual fusion gate (Section 3.5) provides a mild additional benefit. The adapter bottleneck is reverted to 64 (from 128) and the adaptation schedule shortened (40 → 30 epochs), resulting in a slight DA regression to 77.36%. V5's design is only justified when ensemble inference is used: the 0.6:0.4 Main+DA ensemble recovers 84.51% UA, exceeding all single-model results.

Table 4: Adapter bottleneck dimension b vs. DA UA (%). Other architecture differences between versions are noted.

Version	b	DA UA	Note
V3	64	78.29	—
V4	128	**78.66**	+Focal/SA/WR
V5	64	77.36	+dual fus., 30 ep.

Adapter bottleneck size. Table 4 compares bottleneck dimensions across versions. With 864 target utterances, b=128 provides marginally higher capacity than b=64 without overfitting. b=256 (not tabulated) showed signs of overfitting to the small target set. We recommend b=64–128 for datasets of this size.

Main/NA Pareto trade-off. A consistent Pareto frontier emerges: configurations that specialize the model to the source domain (e.g., partial backbone unfreeze in V5) improve Main UA and reduce its variance but degrade zero-shot NA UA. Practitioners should choose V4 when zero-shot deployment is required and V5-Ensemble when a small target adaptation set is available.

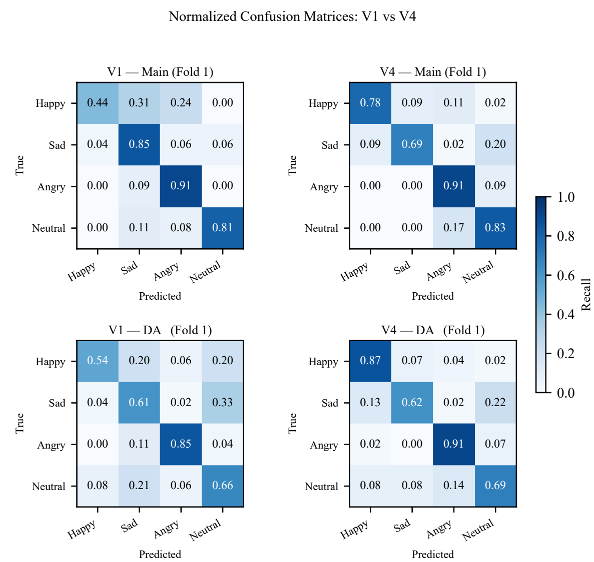

Figure 4: Normalized confusion matrices. Top row: Main protocol (train from scratch on BANGLASER fold 1). Bottom row: DA protocol (adapter fine-tuned from SUBESCO). Left column: V1 (baseline, layer-unfreezing). Right column: V4 (cross-attention + adapters + Focal loss). V4 strongly reduces happy↔sad confusions and improves the neutral and sad diagonal (the two upweighted classes).

7. Analysis

Confusion matrix analysis. Figure 4 shows normalized confusion matrices for V1 and V4 on BANGLASER fold 1 under both the Main and DA protocols. In V1-Main, the most prominent off-diagonal cell is happy misclassified as sad (31%), suggesting that the model conflates high-energy positive arousal with low energy. V4-Main reduces this error to 9% through the combined effect of class-weighted Focal loss (which amplifies the gradient for misclassified sad and neutral samples) and cross-attention fusion (which allows the model to leverage pitch contour via prosody-informed query tokens). The happy diagonal improves from 44% (V1-Main) to 78% (V4-Main), while neutral recall remains strong at 83% (V4-Main, up from 81% in V1-Main). Under the DA

Table 5: Per-class macro-F1 (Main protocol) computed from aggregated confusion matrices across all 5 folds. Sad and happy are the hardest classes throughout; Focal loss with class weights (V4+) narrows the gap.

Model	Happy	Sad	Angry	Neutral
V1	0.622	0.671	0.770	0.808
V2	0.648	0.654	0.797	0.762
V3	0.739	0.662	0.835	0.795
V4	0.773	0.738	**0.855**	0.831
V5	**0.788**	**0.748**	0.879	**0.862**

protocol, V1 shows a more diffuse confusion matrix—a consequence of backbone perturbation during adaptation that disrupts learned feature boundaries (e.g., happy recall drops to 54%, neutral to 66%). V4-DA achieves substantially cleaner separation, with happy recall at 87% and angry at 91%.

Per-class F1 analysis. Table 5 reports per-class F1 from the Main protocol, computed from the aggregated confusion matrices across all five folds. Happy and sad are consistently the weakest classes, while angry is the easiest. Focal loss with class weights $[1.0, 1.3, 1.0, 1.2]$ (introduced in V4) produces the largest single-version gain for sad (F1: $0.662 \rightarrow 0.738$, $+0.076$) and happy (F1: $0.739 \rightarrow 0.773$, $+0.034$), confirming that upweighting underperforming classes is effective. V5 further improves all classes, with neutral reaching 0.862.

Table 6: Per-fold UA (%) for V5-Main, V5-DA, and their ensemble. The ensemble recovers accuracy in folds where one model fails.

Fold	V5-Main	V5-DA	V5-Ensemble
Fold 1	83.89	75.00	82.78
Fold 2	86.57	79.86	81.25
Fold 3	78.06	77.78	83.89
Fold 4	82.83	65.72	82.95
Fold 5	82.41	88.43	91.67
Mean ± std	82.75±2.76	77.36±7.35	84.51±3.68

Ensemble complementarity. Table 6 shows per-fold results for V5. The DA model underperforms Main in fold 4 (early stopping triggered at epoch 3 due to small adaptation set size), while both models perform comparably in folds 2 and 3. The ensemble consistently matches or outperforms either single model, confirming that Main and DA predictions are partially complementary. In fold 4, where the DA model degrades to 65.72%, the ensemble (82.95%) substantially recovers performance by downweighting the DA logits (weight 0.4 vs. 0.6 for Main). In fold 5, DA peaks at 88.43% and the ensemble reaches 91.67%, the highest per-fold UA in our study. This suggests that the 0.6:0.4 blend is a robust operating point that hedges against DA adaptation failures caused by small per-fold target speaker counts.

Table 7: Paired t-tests (df=4) across 5 folds for key comparisons. $|t| > 2.78$ corresponds to $p < 0.05$ (two-tailed).

Comparison	Protocol	t	ΔUA	Sig.
V1 → V4	NA	−5.61	+15.7	✓
V1 → V4	DA	−5.00	+17.4	✓
V1 → V4	Main	−1.91	+7.7	—
V3 → V4	NA	−2.52	+3.1	—
V3 → V4	DA	−0.12	+0.4	—

Statistical significance. Table 7 reports paired t-tests across the five speaker-independent folds for key version comparisons. The central claim of the paper—that adapter-based DA outperforms layer-unfreezing (V1 → V4)—is statistically significant for both the NA protocol (t=−5.61, p<0.01) and the DA protocol (t=−5.00, p<0.01). The Main protocol improvement (+7.7 UA) does not reach significance with only five folds, which is expected given the higher per-fold variance in Main training. The V3 → V4 regularization gains are modest and not significant at α=0.05, consistent with their incremental nature.

Learned block weights. At convergence, the learnable block weights $\{w_\ell\}$ (Equation 1) in V4 assign the largest weight to block 9 ($w_9 \approx 0.38$) and the smallest to block 1 ($w_1 \approx 0.08$), consistent with the analysis of Pasad et al. (2021) who showed that middle and upper blocks (blocks 6–12) are more informative for emotion tasks. Blocks 4, 7, and 12 receive intermediate weights (≈0.15–0.20). Notably, the top block (12) receives a non-trivial weight despite its high-level semantic focus, suggesting it contributes complementary information beyond the prosodic-emotion discriminative signal of block 9.

Limitations. Our study has three main limitations. First, the target corpus is small (864 utterances), which amplifies per-fold variance and makes per-speaker analysis unreliable. Second, we evaluate only four emotions; additional corpora covering other Bangla emotional categories (e.g., fear, disgust) are needed to validate generality. Third, V3's simultaneous introduction of cross-attention and adapters prevents clean attribution of their independent contributions; future work should ablate these individually.

8. Conclusion

We presented the first systematic cross-corpus speech emotion recognition study for Bangla, adapting models trained on SUBESCO to BANGLASER across three evaluation protocols. Our central contribution is demonstrating that *frozen-backbone bottleneck adapters* are dramatically more effective than layer-unfreezing for cross-corpus domain adaptation: replacing the adaptation strategy of the AFTL baseline yields a +16.2 UA improvement under identical data conditions (V2-DA: 62.08% $\rightarrow$ V3-DA: 78.29%). We attribute this to the fact that adapter-based adaptation preserves the pretrained acoustic representations that are essential for cross-corpus generalization, while layer-unfreezing perturbs and erases them.

Complementing the adapter finding, prosody-guided cross-attention with four parallel query tokens provides richer prosody-acoustic alignment than simple multiplicative gating, contributing to improved performance across all three protocols in V3. Progressive regularization via Focal loss, SpecAugment, and cosine warm-restart scheduling (V4) further improves all metrics and establishes the best single-model configuration: 80.52% Main UA, 63.59% zero-shot UA, and 78.66% DA UA. The final V5 ensemble achieves **84.51% UA** on BANGLASER, the best reported result on this corpus.

The key practical insight from our ablation is a Pareto trade-off between source specialization and cross-corpus generalization: partial backbone unfreeze during source training improves Main UA (+2.2 UA, lower variance) but substantially hurts zero-shot transfer (-10.3 UA). This suggests that practitioners should choose V4 when zero-shot deployment is required and the V5 ensemble when a small target adaptation set is available.

Future work includes: (1) extending the evaluation to additional Bangla emotional speech corpora and emotion categories; (2) ablating cross-attention and adapters independently to resolve the V3 confound; (3) comparing bottleneck adapters with LoRA-based adaptation; (4) applying our adapter-based CCSER approach to other low-resource South and Southeast Asian languages where self-supervised speech models have been pre-trained on limited data.

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A. Appendix

A.1. Hyperparameter Details

Table 8 lists all training hyperparameters. Values marked with "*" differ between V5 and V1–V4.

Table 8: Training hyperparameters.

Hyperparameter	Source	Main	DA
Max epochs	80	100	30
Early stopping patience	15	20	8
Optimizer	Adam		
Learning rate	10^{-3}	10^{-3}	5×10^{-5}
W2V2 LR (V5 partial unfreeze)	—	10^{-5}	—
Weight decay	10^{-4}	10^{-4}	10^{-4}
Batch size	8	8	8
AM-Softmax margin m	0.4	0.4	—
AM-Softmax scale s	30	30	—
Focal γ (V4/V5)	2.0	2.0	—
Class weights (V4/V5)	[1.0, 1.3, 1.0, 1.2]		—
SpecAugment (V4/V5)	2T+2F	2T+2F	—
Cosine restart T_0 (V4/V5)	10	10	—
Cosine restart T_{mult}	2	2	—
Loss α	1.0	1.0	—
Loss β	100.0	100.0	—
Adapter bottleneck b	—	—	V3:64, V4:128, V5:64
Ensemble weights	—	—	0.6:0.4
Audio length (s)	7.0 (zero-padded)		
Sample rate (kHz)	16		
W2V2 blocks	{1} (V1), {1,4,7,9,12} (V2–V5)		
Embedding dim	128		
Cross-attn heads	4 (V3+)		
Self-attn heads	8		
Conv1d (channels, k, s)	768→64, k=7, s=3		
Prosody MLP	103→256→64		
Embedding dropout	0.3		

A.2. Prosody Feature Composition

The 103-dimensional prosody vector is extracted from each utterance using Parselmouth (version 0.4.3) and comprises:

- **F0 statistics** (12 values): mean, std, min, max, 10th/25th/50th/75th/90th percentiles, mean absolute slope, voiced frame count, voiced frame ratio.
- **Energy statistics** (12 values): same set of statistics applied to RMS energy.
- **Duration** (1 value): utterance duration in seconds.
- **Jitter** (5 values): local, local absolute, rap, ppq5, ddp.
- **Shimmer** (6 values): local, local dB, apq3, apq5, apq11, dda.
- **HNR** (1 value): harmonics-to-noise ratio.
- **Formants F1–F3** (18 values): mean and std for each of the first three formants, plus bandwidth statistics.
- **Spectral tilt and balance** (6 values): Hammarberg index, spectral tilt, alpha ratio, beta ratio, level (dB SPL), spectral slope.
- **MFCC statistics** (42 values): mean and std of 21 MFCC coefficients (Parselmouth MFCC, 12 coefficients + log energy).

All features are utterance-level global statistics (not framewise); they are normalised to zero mean and unit variance using training-fold statistics.

A.3. Per-Fold Results (V4 and V5)

Table 9 and Table 10 report per-fold UA for V4 and V5 respectively, enabling reproducibility checks.

Table 9: V4 per-fold UA (%).

Fold	Main Ep.	Main	NA	DA Ep.	DA
Fold 1	10	80.28	56.53	15	77.36
Fold 2	5	82.18	67.36	10	85.42
Fold 3	5	72.64	55.69	3	70.97
Fold 4	4	82.32	70.83	5	77.59
Fold 5	5	85.19	67.55	17	81.94
Mean±std	–	80.52±4.24	63.59±6.24	–	78.66±4.87

Table 10: V5 per-fold UA (%).

Fold	Main	NA	DA Ep.	DA	Ensemble
Fold 1	83.89	51.81	14	75.00	82.78
Fold 2	86.57	52.55	26	79.86	81.25
Fold 3	78.06	49.17	25	77.78	83.89
Fold 4	82.83	57.01	3	65.72	82.95
Fold 5	82.41	56.00	20	88.43	91.67
Mean±std	82.75±2.76	53.30±2.86	–	77.36±7.35	84.51±3.68

A.4. Source Domain (SUBESCO) Results

Table 11 reports 5-fold cross-validation results on the source corpus SUBESCO, demonstrating that our architectural improvements also benefit in-domain recognition.

Table 11: Source domain (SUBESCO) 5-fold UA (%).

Model	UA	WA
V1 (AFTL-style)	72.79±5.84	72.82±5.91
V2	73.20±5.79	73.20±5.79
V3	75.43±3.21	75.12±3.45
V4	79.86±3.87	79.42±4.01
V5	81.93±2.44	81.55±2.61

A.5. Comparison with AFTL Baseline Numbers

The original AFTL paper (Naderi and Nasersharif, 2023) reports results on IEMOCAP (English) as source corpus, tested on EMODB (German, 92.45%), ShEMO (Persian, 88.47%), TESS (English, 84.56%), SAVEE (English, 67.48%), and EMOVO (Italian, 66.54%). These numbers are *not* directly comparable to ours as they use different source and target corpora. Our paper is the first to evaluate any CCSER method on the SUBESCO→BANGLASER pair, establishing the baseline numbers reported in Table 2.